Number 7 — Recurring decimals · Advanced calculator problems

Question 1 and 4 without a calculator. Marks in brackets. ★ = harder question. Dot notation: $0.\dot{2}\dot{7} = 0.272727\ldots$

1 Convert each recurring decimal to a fraction in its simplest form. [12]

a $0.\dot{7}$ **b** $0.\dot{2}\dot{7}$ **c** $0.\dot{4}2\dot{3}$ **d** $0.0\dot{4}\dot{5}$ **e** $0.8\dot{3}$ **f** $2.1\dot{6}$

Let x be the decimal; multiply by 10, 100 or 1000 (one, two or three repeating digits) and subtract so the repeating part cancels.

a $10x - x = 7.\dot{7} - 0.\dot{7} = 7 \Rightarrow x = \dfrac{7}{9}$

b $100x - x = 27 \Rightarrow x = \dfrac{27}{99} = \dfrac{3}{11}$

c $1000x - x = 423 \Rightarrow x = \dfrac{423}{999} = \dfrac{47}{111}$

d $100x = 4.\dot{4}\dot{5}$ and $x = 0.0\dot{4}\dot{5}$: $99x = 4.5 \Rightarrow x = \dfrac{4.5}{99} = \dfrac{45}{990} = \dfrac{1}{22}$

e $10x = 8.\dot{3}$, $100x = 83.\dot{3}$: $90x = 75 \Rightarrow x = \dfrac{75}{90} = \dfrac{5}{6}$

f $10x = 21.\dot{6}$, $100x = 216.\dot{6}$: $90x = 195 \Rightarrow x = \dfrac{195}{90} = \dfrac{13}{6} = 2\tfrac{1}{6}$

2 Calculate, giving each answer correct to 3 significant figures. [6]

a $\dfrac{1}{2^3} + \dfrac{1}{3^3} + \dfrac{1}{4^3}$ **b** $\dfrac{6.4 \times 10^4}{8.15 \times 10^6}$ **c** $\sqrt[5]{\dfrac{7.2 \times 10^4}{3.1 \times 10^{-2}}}$

a $0.125 + 0.037037 + 0.015625 = 0.178$. **b** $0.007853\ldots = 7.85 \times 10^{-3}$. **c** Inside the root: 2.32×10^6; fifth root (use the $\sqrt[x]{}$ key or power $\frac{1}{5}$): 18.8.

3 Find the value of z to 3 significant figures, given that $z = \sqrt[3]{\dfrac{a^2 + b^2 + c^2}{\tan(b - c)^\circ}}$ where $a = 40$, $b = 50$ and $c = 20$. [4]

Numerator $1600 + 2500 + 400 = 4500$; denominator $\tan 30^\circ = 0.5774$; quotient 7794.2; cube root: $z = 19.8$. Keep the full calculator value until the final rounding.

4 a and b are digits. Show that $1.\dot{a}\dot{b} = \dfrac{ab + 99}{99}$, where ab is the two-digit number with digits a and b. [3]

> Let $N = 1.abab\ldots$ Two digits repeat, so multiply by 100: $100N = 1ab.abab\ldots$ Subtracting, $99N = 1ab - 1$. The three-digit number $1ab$ is $100 + ab$, so $99N = 100 + ab - 1 = ab + 99$ and $N = \dfrac{ab + 99}{99}$. (Check with $a = 2$, $b = 7$: $\dfrac{126}{99} = 1.2727\ldots$ ✓)

Total: 25 marks

Check: one repeating digit → ×10, two → ×100, three → ×1000, then subtract · a non-repeating digit after the point: multiply by 10 first, then by 10 again (×100) and subtract · always simplify the fraction · calculator: brackets round numerators and denominators, keep full accuracy, round at the end.